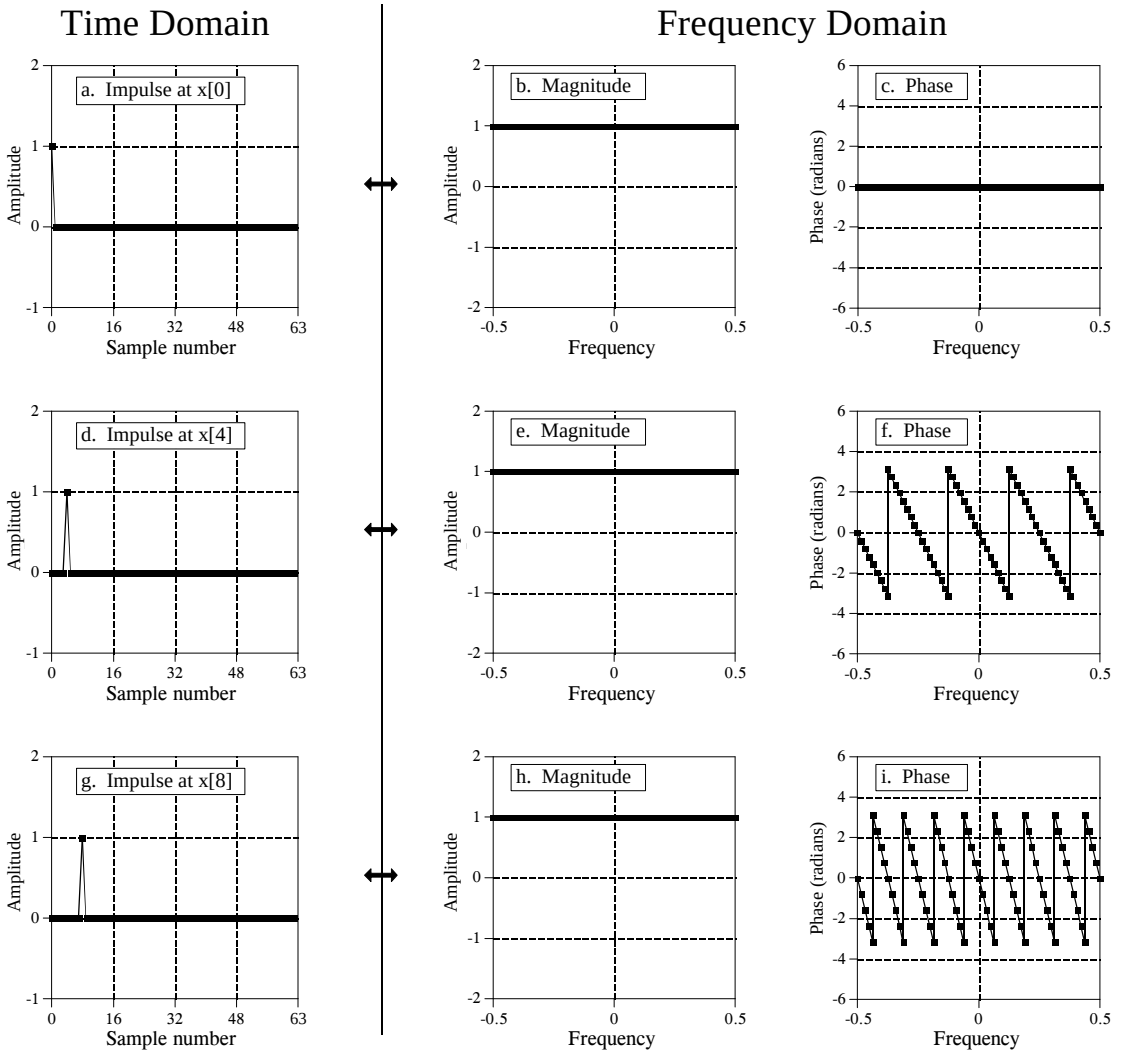

FIGURE 11-1
Delta function pairs in *polar form*. An impulse in the time domain corresponds to a constant magnitude and a linear phase in the frequency domain.

frequency domains. As is usually the case, the polar form is much easier to understand; the magnitude is nothing more than a constant, while the phase is a straight line. In comparison, the real and imaginary parts are sinusoidal oscillations that are difficult to attach a meaning to.

The second interesting feature in Fig. 11-2 is the *duality* of the DFT. In the conventional view, each sample in the DFT's frequency domain corresponds to a sinusoid in the time domain. However, the reverse of this is also true, each sample in the time domain corresponds to sinusoids in the frequency domain. Including the negative frequencies in these graphs allows the duality property to be more symmetrical. For instance, Figs. (d), (e), and